Technology Justification & Architectural Trade-offs

Course: DBMS Mini Project (UE24CS395) • **Institution:** PES University (2026–2027)
Team: Saatvik Gupta (PES1UG24CS395) & Satyam Goyal (PES1UG24CS424)

1. The Polyglot Persistence Requirement

Modern law enforcement fusion centers process two divergent classes of data that cannot be satisfied by any single database model:
1. **Strict Relational Entities (PostgreSQL / SQLite):** Legal chain of custody, evidence accounting, and officer roles require strict **ACID transactions**, foreign key referential integrity, and cascading rules. Storing court exhibits in an eventually consistent NoSQL store risks race conditions and evidence being ruled inadmissible in court.
2. **Polymorphic Forensic Artifacts (MongoDB):** Intelligence feeds (ANPR camera telemetry, CCTV frame EXIF, phone extraction JSON, wiretap logs) possess unpredictable, evolving schemas. In a relational database, accommodating these would require continuous schema migrations (ALTER TABLE) or loss of queryability in unstructured text blobs. MongoDB naturally models these as polymorphic BSON documents.

2. Architectural Trade-off Matrix

Parameter	PostgreSQL (Relational Core)	MongoDB (Document Store)
Primary Role	System of Record for Entities & Custody	Elastic Intelligence Document Repository
Integrity Guarantees	Full ACID, Serialized Transactions	Document-level atomicity, schema agility
Relationships	Foreign Keys & 12 Junction Tables (M:N)	Polymorphic application-level references
Search & Retrieval	Exact B-tree lookups, Joins, Window Functions	Inverted Text Index, 64-Dim Vector Similarity
Procedural Logic	PL/pgSQL Procedures, Functions, Triggers	Multi-stage Aggregation Pipelines ($group, $sort)

3. Application Stack Rationale (FastAPI & Next.js 15)

• **FastAPI:** Asynchronous Python framework federating PostgreSQL/SQLite queries and MongoDB calls in parallel, generating automatic OpenAPI/Swagger documentation.
• **Next.js 15 & React:** Server-side rendered interface designed with a dark graphite case-management theme, dense tabular layouts, and zero decorative bloat.